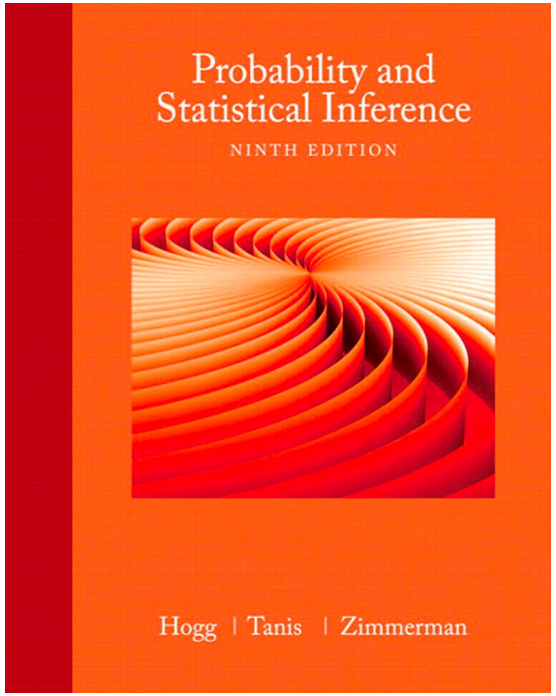
STA 2002, Summer 2023
Probability and Statistics II

Descriptive statistics

Reading

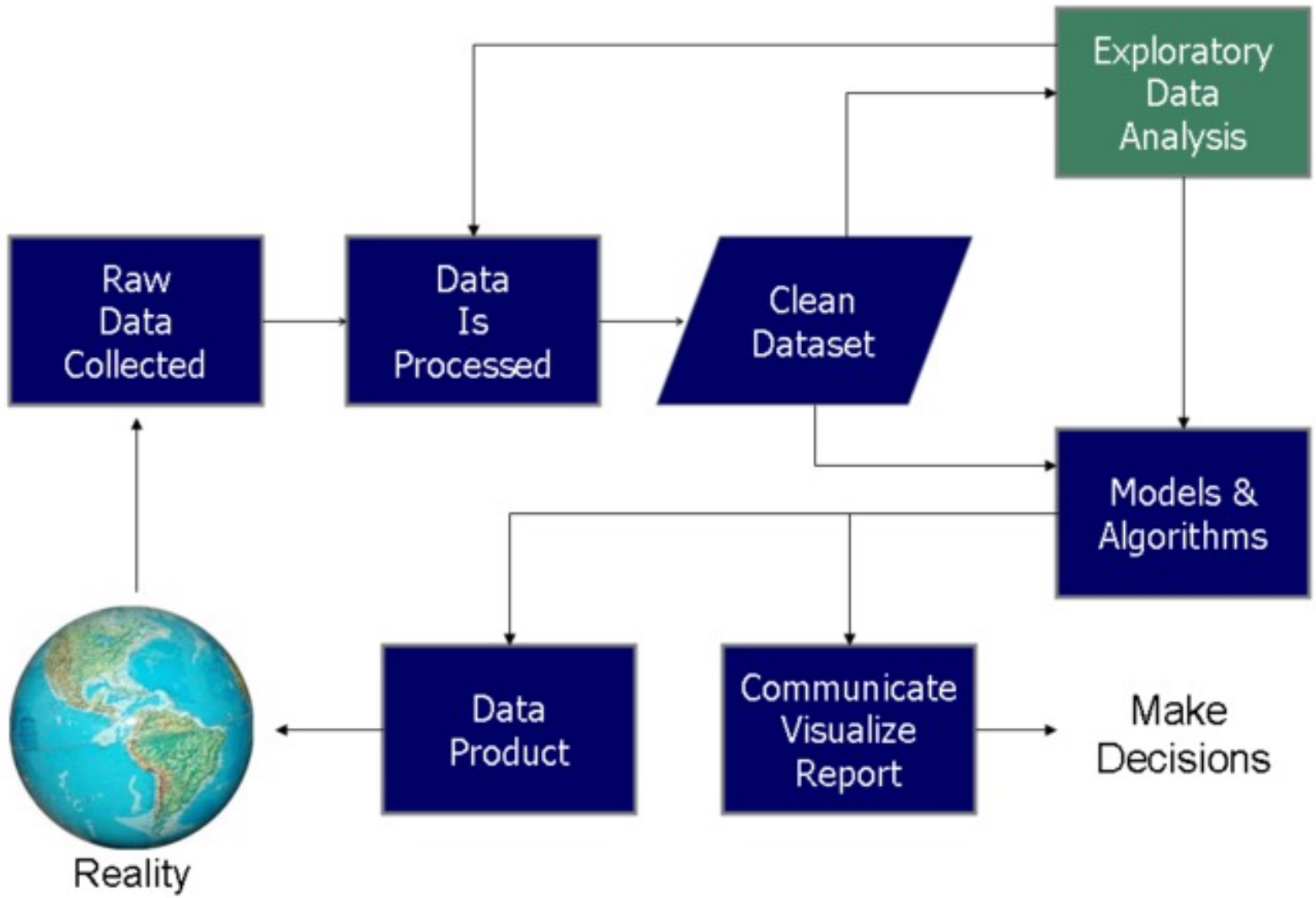
- **Chapter 6.1 & 6.2 in the textbook**



Descriptive statistics

- An important aspect of dealing with data is organizing and summarizing the data in ways that facilitate its interpretation and subsequent analysis.
- This aspect of statistics is called descriptive statistics. It is a summary statistic that quantitatively describes or summarizes features of a collection of information.
- It is usually the first-attempt of data analytics.
- Sometimes they are sufficient for particular investigations.

Data Science Process



Types of Summary Statistics

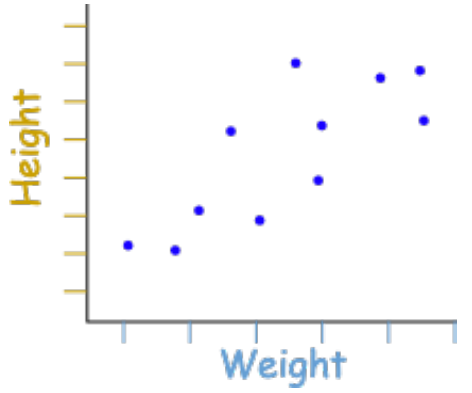
Numerical Summaries

Univariate

- Central tendency (mean, median, mode)
- Variability (dispersion): variance, std, quartiles
- Range: max, min
- Shape: kurtosis and skewness
- Probability plot

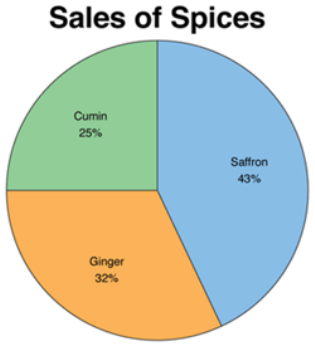
Bivariate

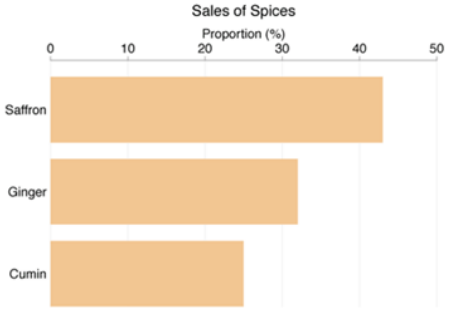
- Scatter plot



Graphical

- Pie chart, bar chart
- Histogram, box plot
- Data visualization

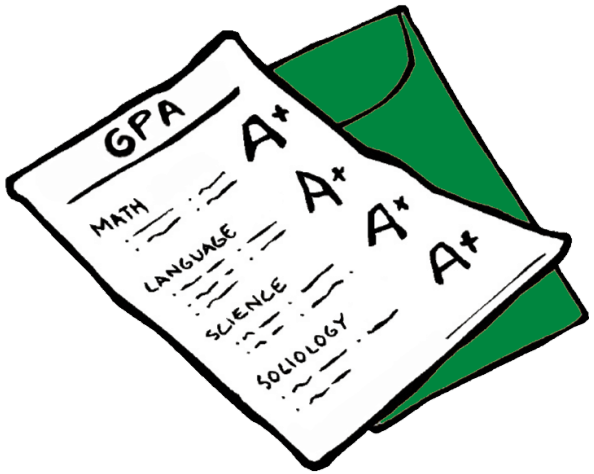




Examples

- **Shooting percentage in basketball is a descriptive statistic that summarizes the performance of a player or a team.**
- **GPA: grade point average. This single number describes the general performance of a student across the range of their course experiences**





Data type

Categorical or nominal data

Observe frequencies within several categories

Example: frequency of female and male attendance in this class

Numerical Data

Observed values are integer, real or complex numbers

Examples: IQ scores of GT students (integer values)

Lifetime of a computer chip (real values)

Class Activity

1. For each CUHKSZ student, we record his/her blood type

A. numeric B. categorical

2. For each CUHKSZ student, we record his/her number of siblings.

A. numeric B. categorical

3. For each CUHKSZ student, we record his/her country of residence.

A. numeric B. categorical

4. For each CUHKSZ student, we record his/her height.

A. numeric B. categorical

Sample Mean

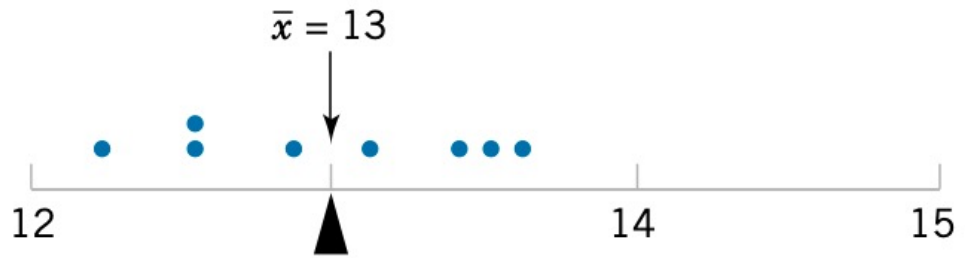
If the n observations in a sample are denoted by $x_1, x_2, \ldots, x_n$, the **sample mean** is

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{n} x_i}{n}$$

For example, the eight observations are

$$x_1 = 12.6, x_2 = 12.9, x_3 = 13.4, x_4 = 12.3, x_5 = 13.6, x_6 = 13.5, x_7 = 12.6, \text{and } x_8 = 13.1.$$

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n} = \frac{\sum_{i=1}^{8} x_i}{8} = \frac{12.6 + 12.9 + \cdots + 13.1}{8} = \frac{104}{8} = 13.0$$



Sample Median

$\tilde{x}$ A value such that 50% of the data are at or above this value

How to calculate:

- Sort the data in ascending (or descending) order
- If *n* is an odd number, median is the $(n+1)/2^{th}$ number
- If *n* is an even number, median is the average of is the $n/2^{th}$ and $(n/2)+1^{th}$ numbers

For example, the eight observations are

$$x_1 = 12.6, x_2 = 12.9, x_3 = 13.4, x_4 = 12.3, x_5 = 13.6, x_6 = 13.5, x_7 = 12.6, \text{and } x_8 = 13.1.$$

What is the median?

Sample Mode

$\hat{x}$ Observation with the highest frequency

For example, the eight observations are

$$x_1 = 12.6, x_2 = 12.9, x_3 = 13.4, x_4 = 12.3, x_5 = 13.6, x_6 = 13.5, x_7 = 12.6, \text{and } x_8 = 13.1.$$

Mode: 12.6

Example 1:

Data = {3, 1, 1, 0, 5, 4, 13, 3}

1. Mean = ?
2. Median = ?
3. Mode = ?

Example 2:

Data = {3, 1, 1, 0, 5, 4, 3}

1. Mean = ?
2. Median = ?
3. Mode = ?

Estimating Data Variability

1. **Sample Range = [Largest item] –[smallest item]**

   Easy to calculate, but often misleading (due to outliers).

If the n observations in a sample are denoted by $x_1, x_2, \ldots, x_n$, the **sample range** is

$$r = \max(x_i) - \min(x_i)$$

2. **Sample Variance measures deviation from the mean**

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n-1}$$

s: Standard deviation

Pay attention to n-1 !

Derivation of the variance formula

$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1} = \frac{\sum_{i=1}^{n} (x_i^2 + \bar{x}^2 + 2\bar{x}x_i)}{n-1} = \frac{\sum_{i=1}^{n} x_i^2 + n\bar{x}^2 - 2\bar{x}\sum_{i=1}^{n} x_i}{n-1}$$

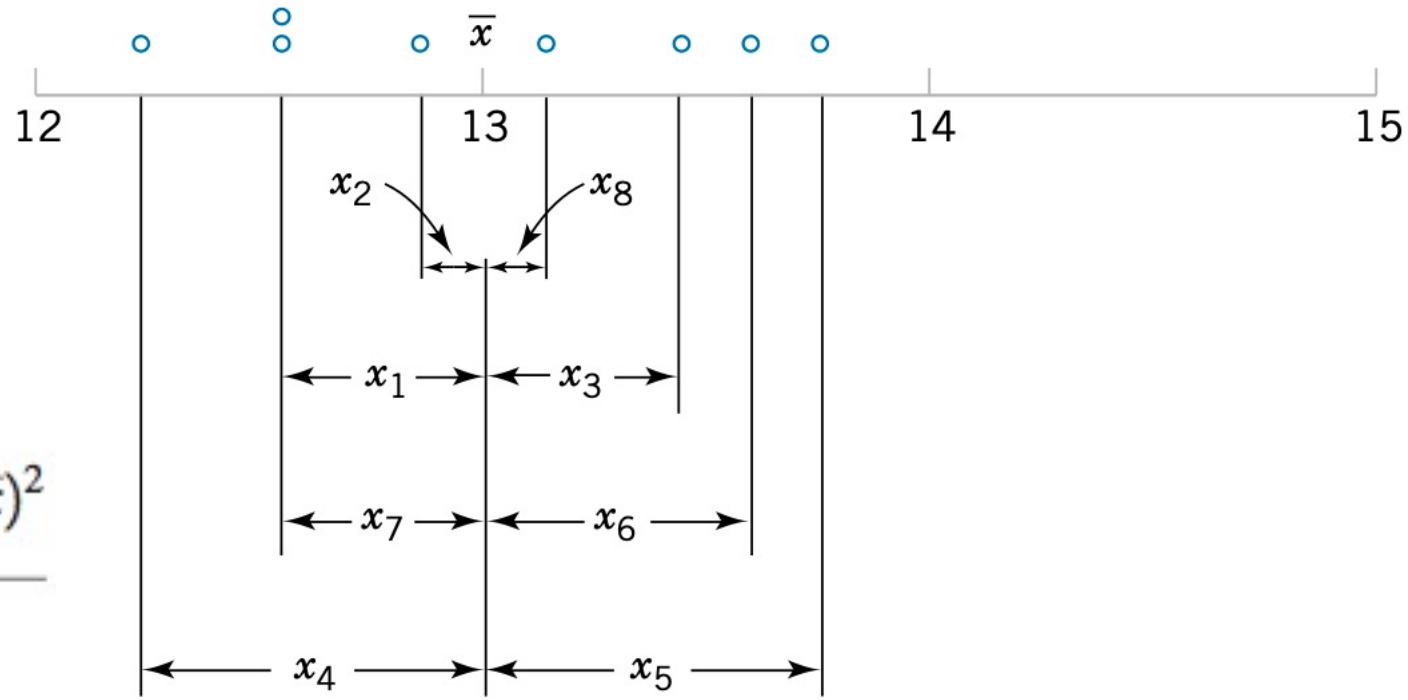
and since $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$, this last equation reduces to

$$s^2 = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n-1}$$

Sample Variance & Standard Deviation

How the sample variance measures variability through the deviations.

$$x_i - \bar{x}$$



$$s^2 = \frac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n - 1}$$

Examples

For example, the eight observations are

$$x_1 = 12.6, x_2 = 12.9, x_3 = 13.4, x_4 = 12.3, x_5 = 13.6,$$
$$x_6 = 13.5, x_7 = 12.6, \text{and } x_8 = 13.1.$$

i	x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	12.6	−0.4	0.16
2	12.9	−0.1	0.01
3	13.4	0.4	0.16
4	12.3	−0.7	0.49
5	13.6	0.6	0.36
6	13.5	0.5	0.25
7	12.6	−0.4	0.16
8	13.1	0.1	0.01
	104.0	0.0	1.60

Sample range

$$r = x_{\max} - x_{\min} = 13.6 - 12.3 = 1.3$$

Sample variance

$$s^2 = \frac{1.60}{8-1} = \frac{1.60}{7} = 0.2286 \quad \text{or} \quad s^2 = \frac{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}{n-1} = \frac{1353.6 - \frac{(104)^2}{8}}{7} = \frac{1.60}{7} = 0.2286$$

std $s = \sqrt{0.2286} = 0.48$

Sample Quantiles

The pth quantile (x_p) is such that 100p% of the sample is smaller than x_p

Hint : Imagine a sample of 100 ranked items. Think of the quantile as an item's **rank**.

Quartile refers to quarters:
25th quantile (Q1, lower-quartile, first-quartile),
50th quantile (Q2, second-quartile, **median**), and
75th percentile (Q3, upper-quartile, third-quartile)

3. Sample inter-quantile range (IQR) = Q3 – Q1

IQR is insensitive to extreme values.

Example:

Data = {3, 1, 1, 0, 5, 4, 13, 3}

1. **Sample Range**

2. **Sample Variance:**

3. **Sample IQR = Upper quartile - Lower quartile**

Data = {3, 1, 1, 0, 5, 4, 3}

Repeat the calculation above.

Numerical Summary of Data

Statistic: Any function of sampled observations is called a statistic

Central Tendency statistics

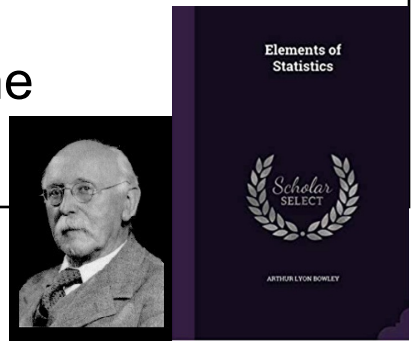
Mean $\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$

Median $\tilde{x}$

A value such that 50% of the data are at or above this value.

Mode $\hat{x}$

Observation with the highest frequency



Arthur Lyon Bowley

Variability statistics

Range $R = x_{\max} - x_{\min}$

Variance $S^2 = \dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}$

Standard Deviation $S = \sqrt{\dfrac{\sum_{i=1}^{n} (x_i - \bar{x})^2}{n-1}}$

IQR $Q_3 - Q_1$

Data Excercise

The data are:

74.001, 74.003, 74.015, 74.000, 74.005, 74.002, 74.005, 74.004

Central Tendency statistics

Mean 74.00437

Median 74.0035

Mode 74.005

Variability statistics

Range 0.015

Variance 2.169643e-05

Standard Deviation 0.004657943

IQR

74.005-74.00175 = 0.00325

Covariance between two variables

- **Given two groups of samples**

$$x_1, x_2, \dots, x_n, \text{ and } y_1, y_2, \dots, y_n$$

- **Sample Covariance:**

$$\frac{1}{n}\left(\sum_{i=1}^{n} x_i y_i\right) - \left(\frac{1}{n}\sum_{i=1}^{n} x_i\right)\left(\frac{1}{n}\sum_{i=1}^{n} y_i\right)$$

- **Estimation of $Cov(X, Y) = E[XY] - E[X]E[Y]$**

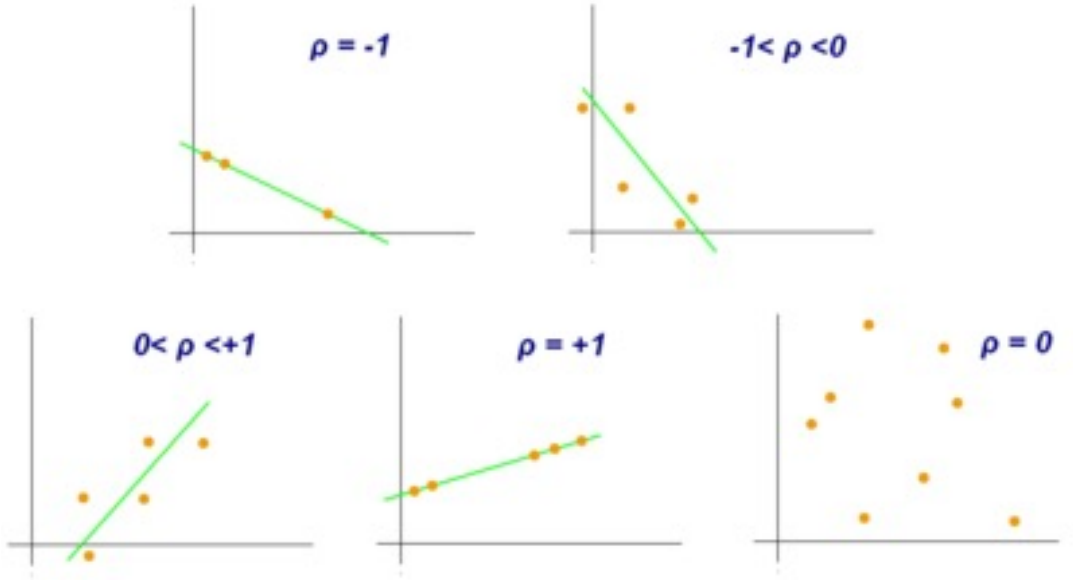
Pearson correlation coefficient

- **Describing the linear correlation between two variables**

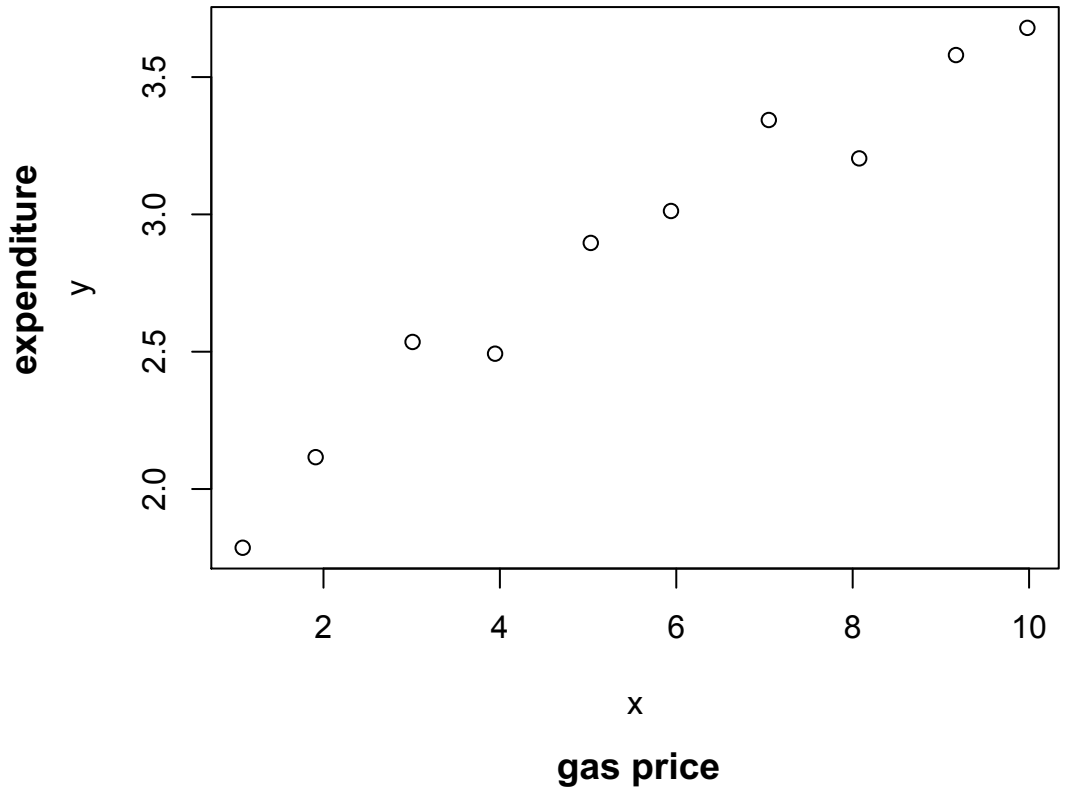
Given two groups of samples

$x_1, x_2, \ldots, x_n$, and $y_1, y_2, \ldots, y_n$

Cor(x,y) $$r_{xy} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$



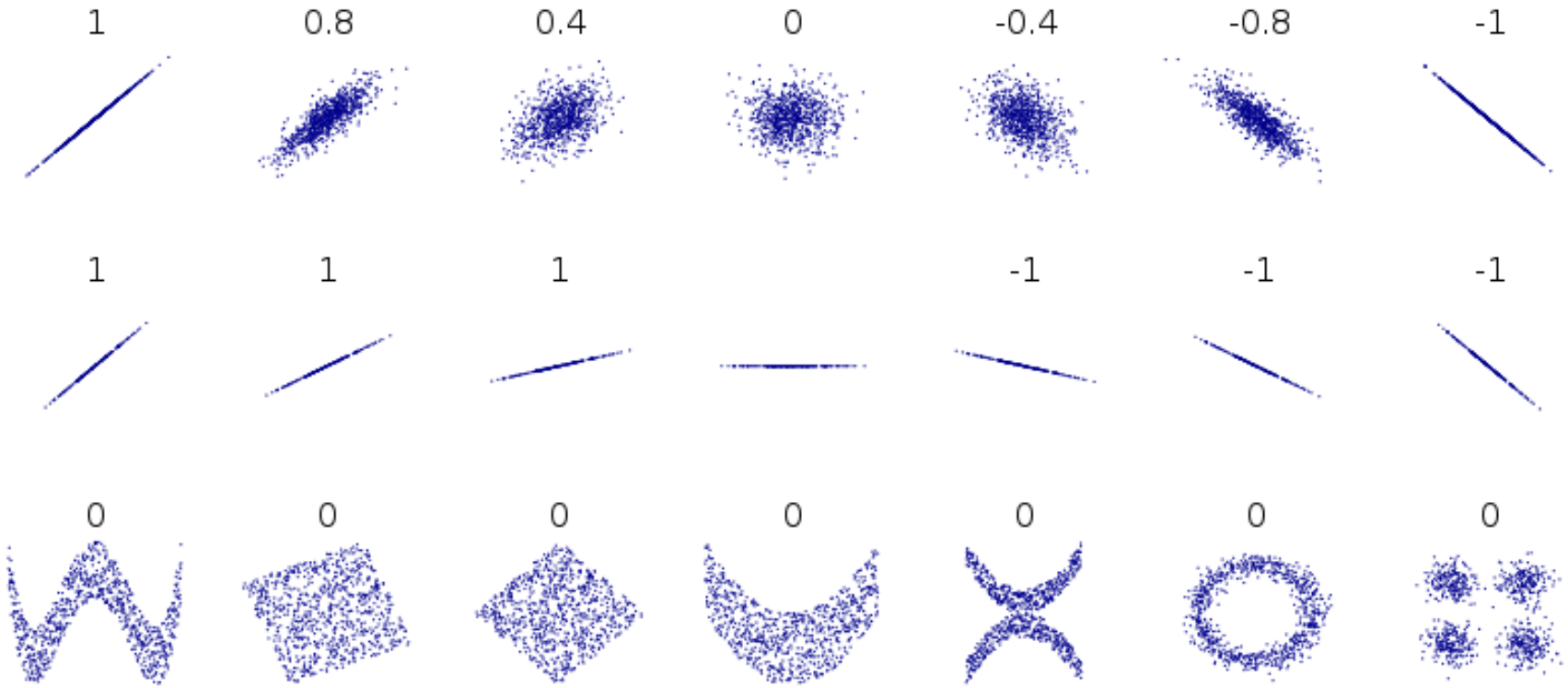
Example



- **Data**
- 1.0840 1.7862
- 1.9112 2.1160
- 3.0100 2.5355
- 3.9455 2.4928
- 5.0304 2.8961
- 5.9400 3.0124
- 7.0490 3.3437
- 8.0739 3.2039
- 9.1712 3.5802
- 9.9806 3.6792

- **cov(x,y) = 1.865434**
- **cor(x,y) = 0.9774152**

What correlation captures

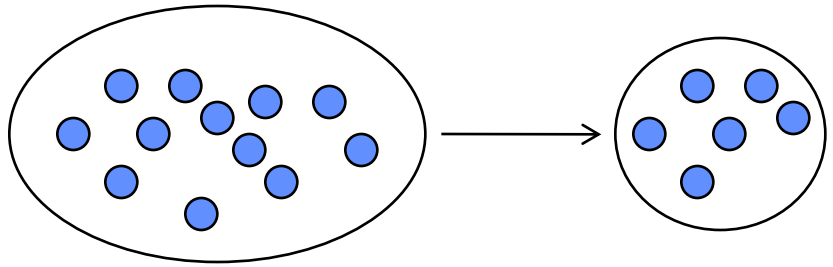


Note that the correlation reflects the non-linearity and direction of a linear relationship (top row), but not the slope of that relationship (middle), nor many aspects of nonlinear relationships (bottom).

Population vs Sample Mean

A **population** consists of the totality of the observations with which we are concerned.

N: population size



n: sample size

population **sample**

A **sample** is a subset of observations selected from a population.

Population mean $\mu = \dfrac{\sum_{i=1}^{N} x_i}{N}$ $\quad \boldsymbol{E[X]}$

$\bar{x} = \dfrac{\sum_{i=1}^{n} x_i}{n}$ **Sample mean**

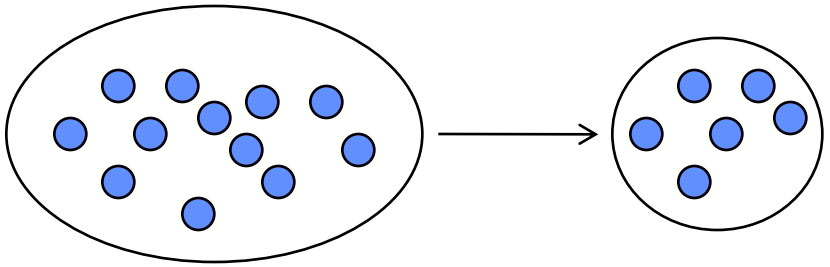
The sample mean is a reasonable estimate of the population mean.

(More accurate when the sample size increases)

Population vs Sample Variance

A **population** consists of the totality of the observations with which we are concerned.

N: population size



population

sample

n: sample size

A **sample** is a subset of observations selected from a population.

Population variance $\sigma^2 = \dfrac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$

$s^2 = \dfrac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$ **Sample variance**

$\boldsymbol{Var[X]}$

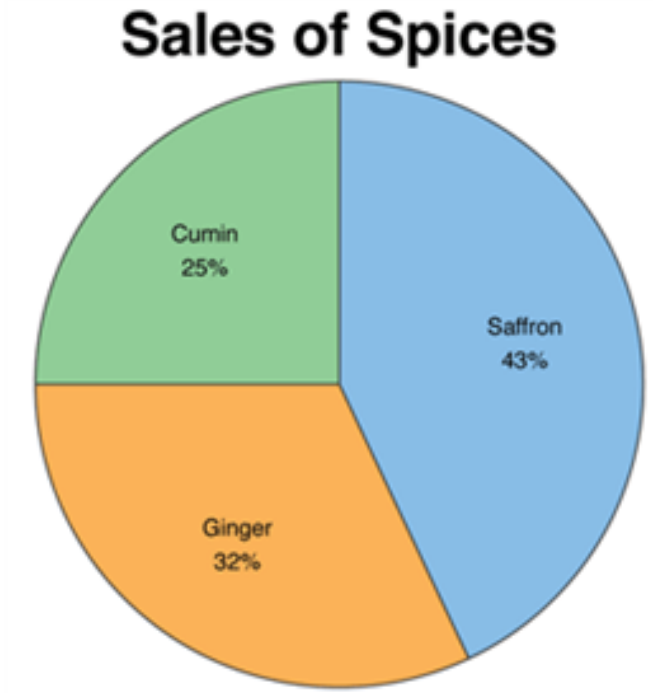
The sample variance is a reasonable estimate of the population variance.

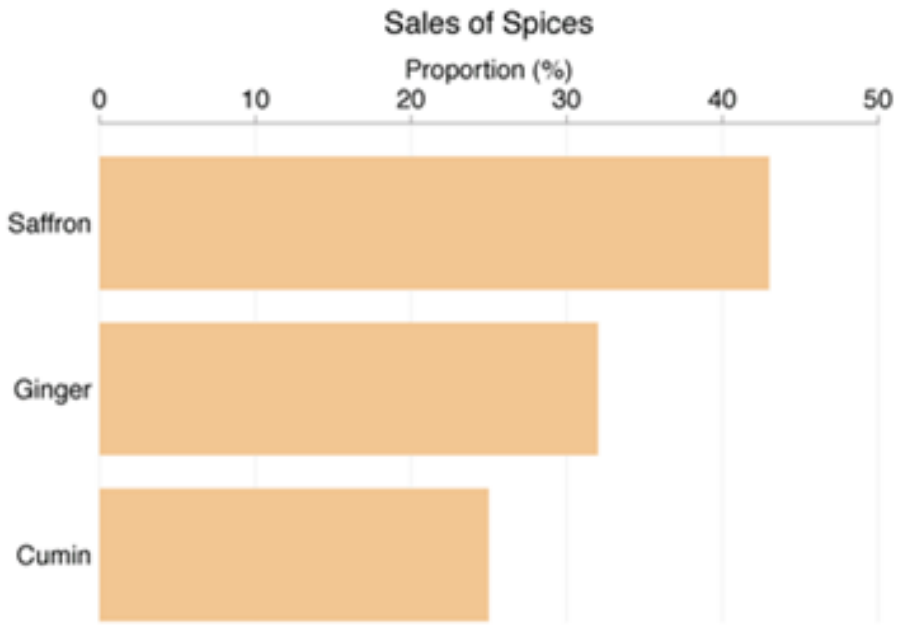
(More accurate when the sample size increases)

Graphical Descriptive Statistics

- Pie chart
- Stem-and-leaf diagram
- Frequency Table and Histogram
- Box plot
- Time-series plot

Pie Chart


Sales of Spices
Cumin
25%
Saffron
43%
Ginger
32%


Sales of Spices
Proportion (%)
0
10
20
30
40
50
Saffron
Ginger
Cumin

Stem-and-Leaf Diagrams

A **stem-and-leaf diagram** is a good way to obtain an informative visual display of a data set $x_1, x_2, \ldots, x_n$, where each number x_i consists of at least two digits. To construct a stem-and-leaf diagram, use the following steps.

Steps for Constructing a Stem-and-Leaf Diagram

(1) Divide each number x_i into two parts: a **stem,** consisting of one or more of the leading digits and a **leaf,** consisting of the remaining digit.

(2) List the stem values in a vertical column.

(3) Record the leaf for each observation beside its stem.

(4) Write the units for stems and leaves on the display.

Stem and Leaf Plots

Data = {33, 28, 16, 35, 11, 44, 33, 38}

(Stem) 1st digit	(Leaf) 2nd digit
0	-
1	1, 6
2	8
3	3, 3, 5, 8
4	4



Time table using a stem-and-leaf layout at Minato Mirai train station in Yokohama, Japan. It is a widespread design pattern in the country.

Stem-and-Leaf Diagrams

Data

105	221	183	186	121	181	180	143
97	154	153	174	120	168	167	141
245	228	174	199	181	158	176	110
163	131	154	115	160	208	158	133
207	180	190	193	194	133	156	123
134	178	76	167	184	135	229	146
218	157	101	171	165	172	158	169
199	151	142	163	145	171	148	158
160	175	149	87	160	237	150	135
196	201	200	176	150	170	118	149



Stem	Leaf	Frequency
7	6	1
8	7	1
9	7	1
10	5 1	2
11	5 8 0	3
12	1 0 3	3
13	4 1 3 5 3 5	6
14	2 9 5 8 3 1 6 9	8
15	4 7 1 3 4 0 8 8 6 8 0 8	12
16	3 0 7 3 0 5 0 8 7 9	10
17	8 5 4 4 1 6 2 1 0 6	10
18	0 3 6 1 4 1 0	7
19	9 6 0 9 3 4	6
20	7 1 0 8	4
21	8	1
22	1 8 9	3
23	7	1
24	5	1

How to display data? Data Features

Shape of the data distribution: symmetric, skewed to the right or to the left.

Spread of the data: range, long or short tails

Outliers: extreme values that appear separate from the rest of the data

Modes: concentrations of the data – unimodal, bimodal, multimodal

Gaps: different subpopulations

Comparison of two or more datasets

Histograms

Divide observations into groups to construct frequency histogram. Often, this is the BEST way to communicate your findings from a set of data.

We can (generally) rely on computers to generate histograms

- **Keep bin widths equal**
- **Choose width that summarizes the data best**
- **Use Excel, R, Minitab, SAS, MATLAB, Python, etc**



Karl Pearson

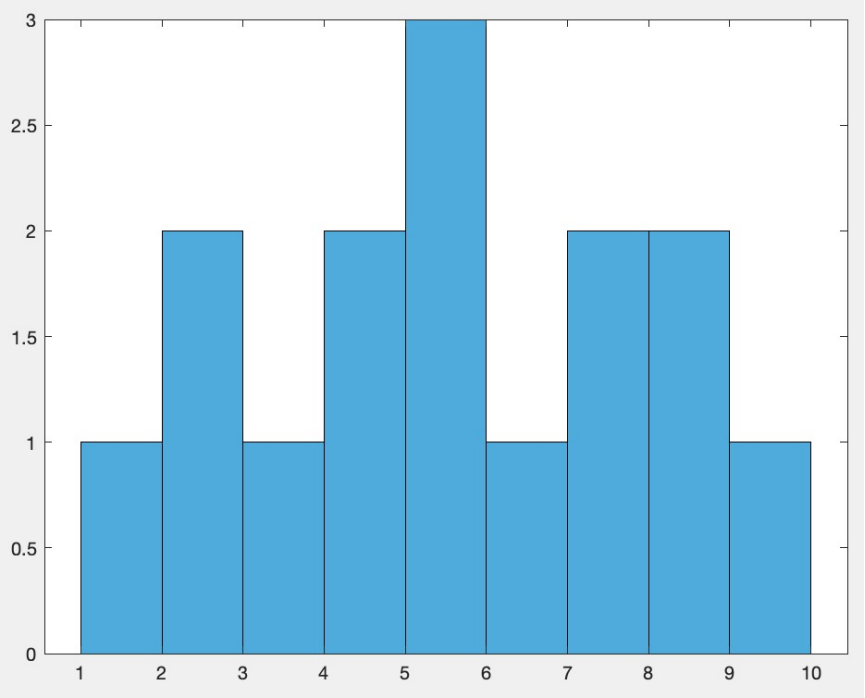
Frequency Table and Histogram

- **To construct a frequency table**
  1. **Find the range of the data**
     - start the upper limit for the last bin just slightly above the largest data value
     - b_0=min(x), b_m>max(x),
     - R = b_m - b_0
  2. **Divide this range into a suitable number *m* of equal intervals**
     - m=4 ~ 20, or $\sqrt{n}$ (n is the total number of samples)
  3. **Count the frequency of each interval**
     - if $b_{i-1} \leq x < b_i$

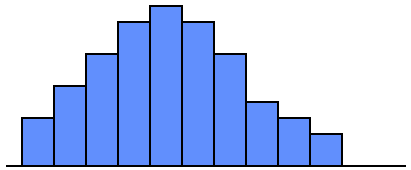
Example

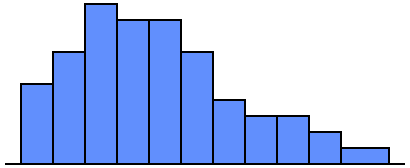
- **Data:**

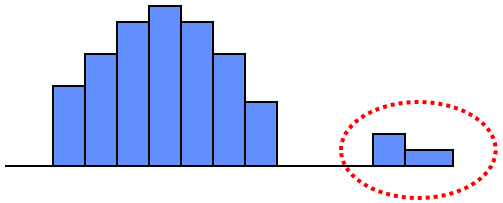
9, 5, 1, 4, 4, 7, 2, 5, 3, 8, 7, 6, 5, 8, 2



Histogram Features







Shape of the data distribution: symmetric, skewed to the right or to the left.

Spread of the data: range, long or short tails

Outliers: extreme values that appear separate from the rest of the data

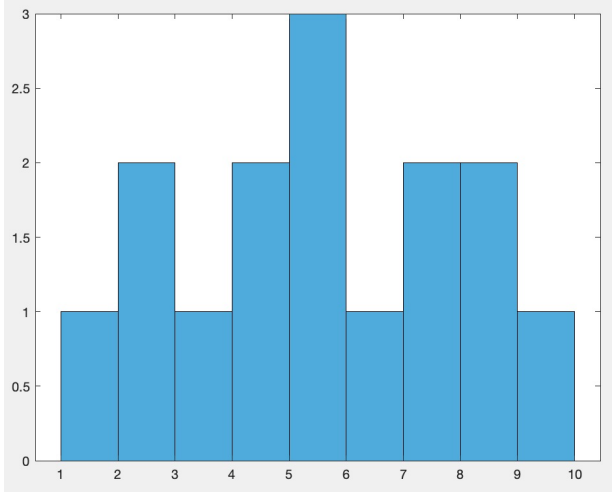
Modes: concentrations of the data – unimodal, bimodal

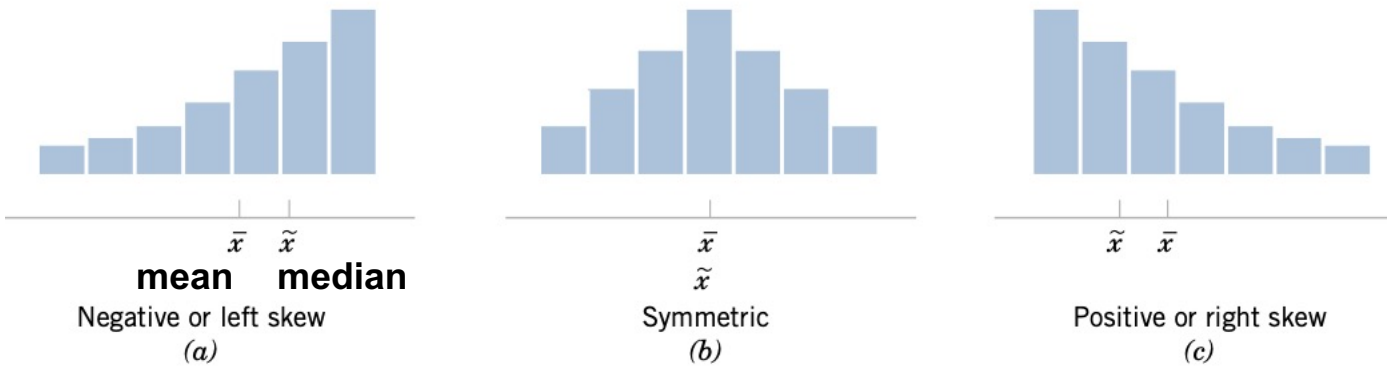
Gaps: different subpopulations

Interpretation based on Histogram

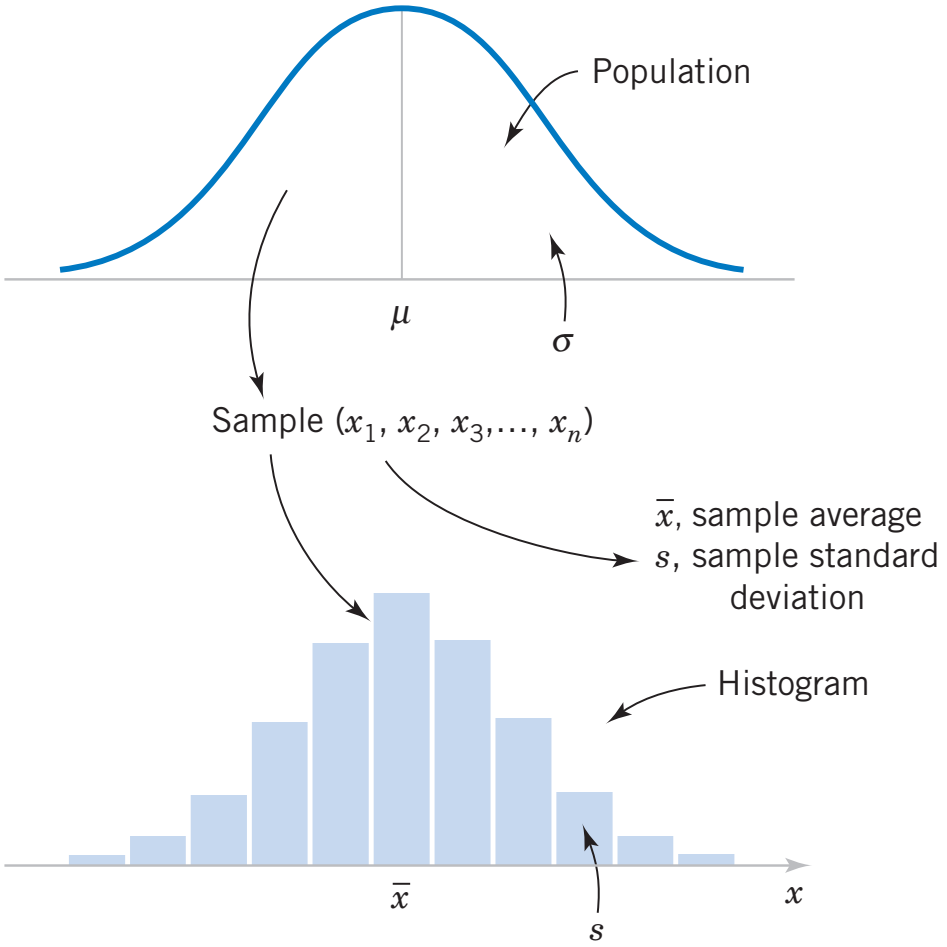
Three Properties of Sample Data

- **Shape:**
  - **roughly symmetric and unimodal**
- **The center tendency or location**
  - **the points tend to cluster near 5.**
- **Scatter or spread range**
  - **variability is relatively high** (min=1; max=9)



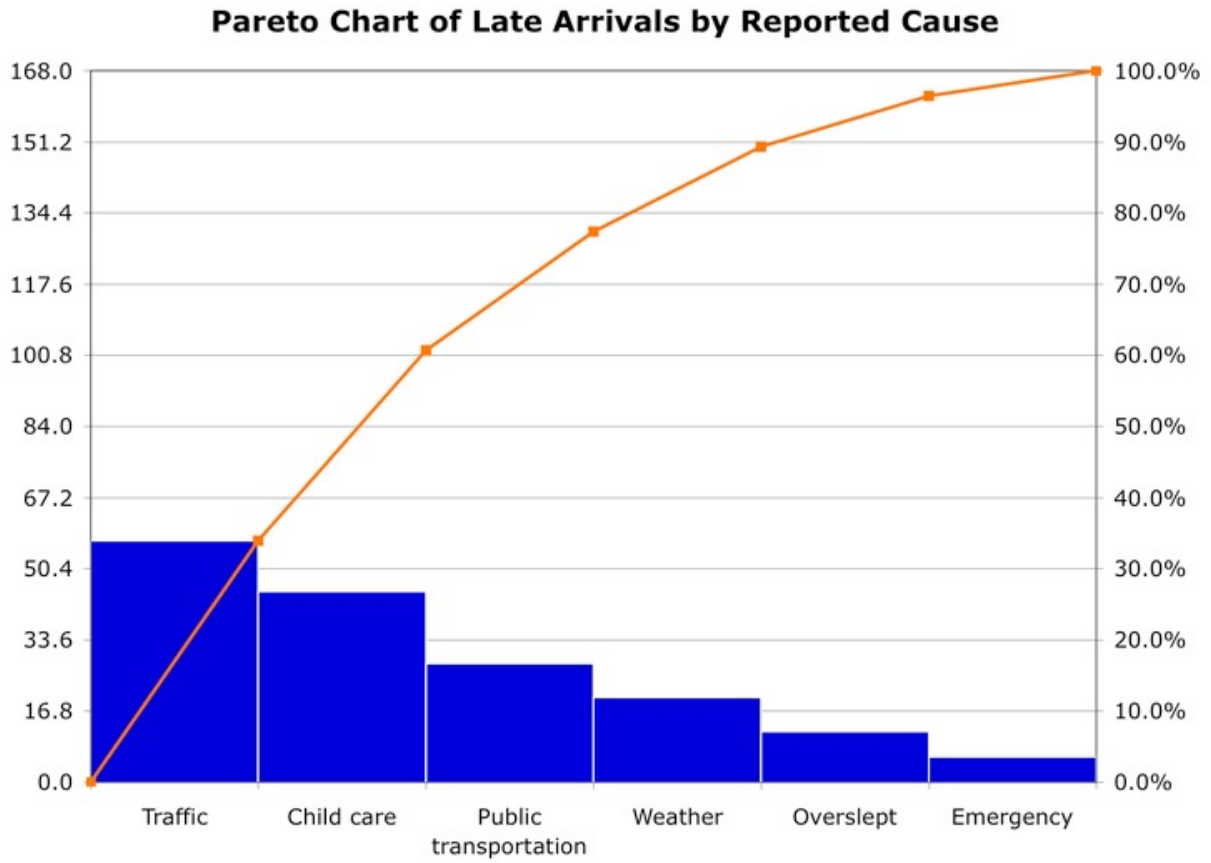


Relationship between population and samples



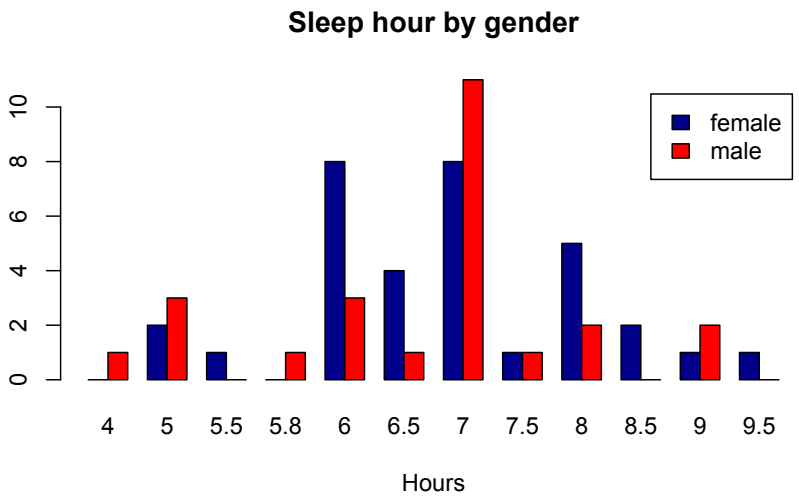
Pareto chart

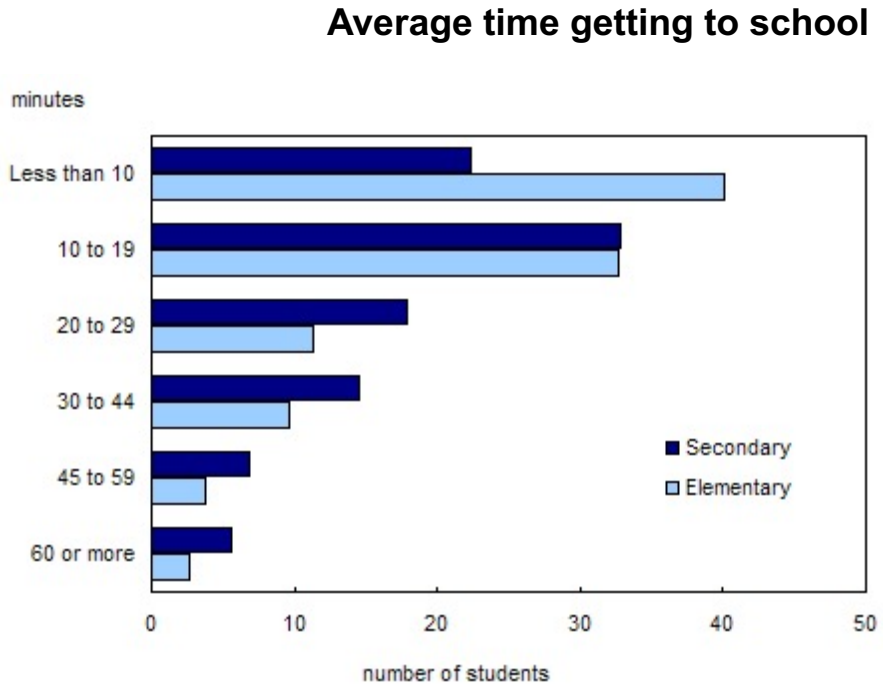
Highlight the most important among a (typically large) set of factors.



Simple example of a Pareto chart using hypothetical data showing the relative frequency of reasons for arriving late at work

Line Graphs: bar chart



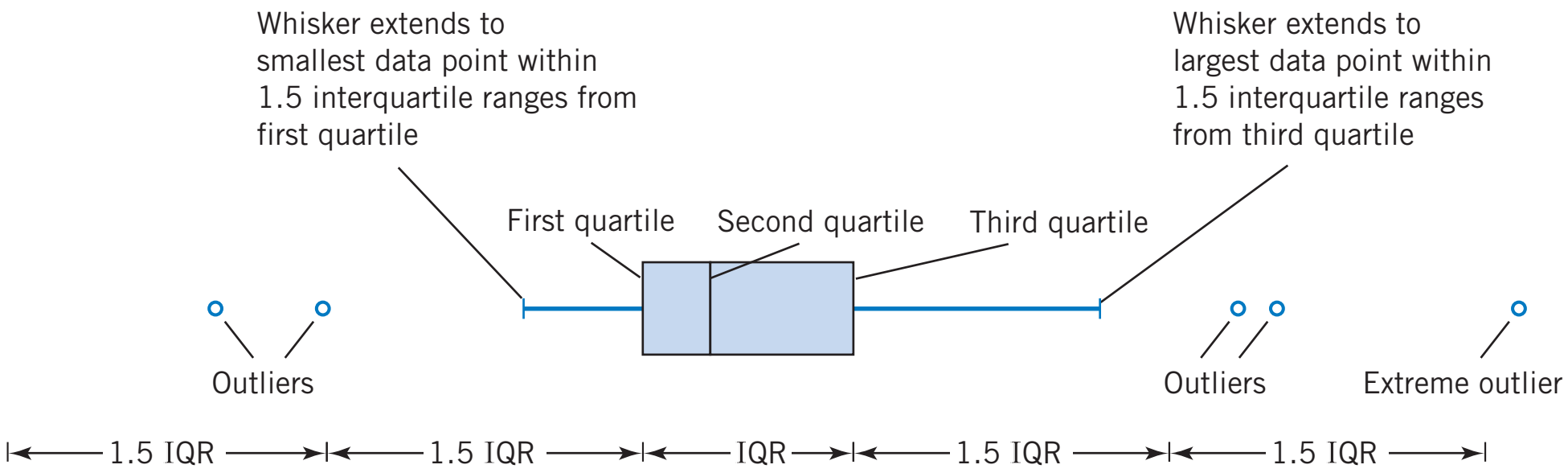


Box Plots

- The box plot is a graphical display that simultaneously describes several important features of a data set, such as center, spread, departure from symmetry, and identification of observations that lie unusually far from the bulk of the data (outliers).

- A box plot displays the **three quartiles**, the **minimum**, and the **maximum** of the data on a rec-tangular box, aligned either horizontally or vertically.
- The box encloses the interquartile range (IQR) with the left (or lower) edge at the first quartile, *q1*, and the right (or upper) edge at the third quartile, *q*3. A line is drawn through the box at the second quartile (which is the 50th percentile or the median), *q*2.
- A line, or whisker, extends from each end of the box. The lower whisker is a line from the first quartile to the smallest data point within 1.5 interquartile ranges from the first quartile. The upper whisker is a line from the third quartile to the largest data point within 1.5 interquartile ranges from the third quartile.
- Data farther from the box than the whiskers are plotted as **individual points**. A point beyond a whisker, but less than 3 interquartile ranges from the box edge, is called an **outlier**. A point more than 3 interquartile ranges from the box edge is called an **extreme outlier**.

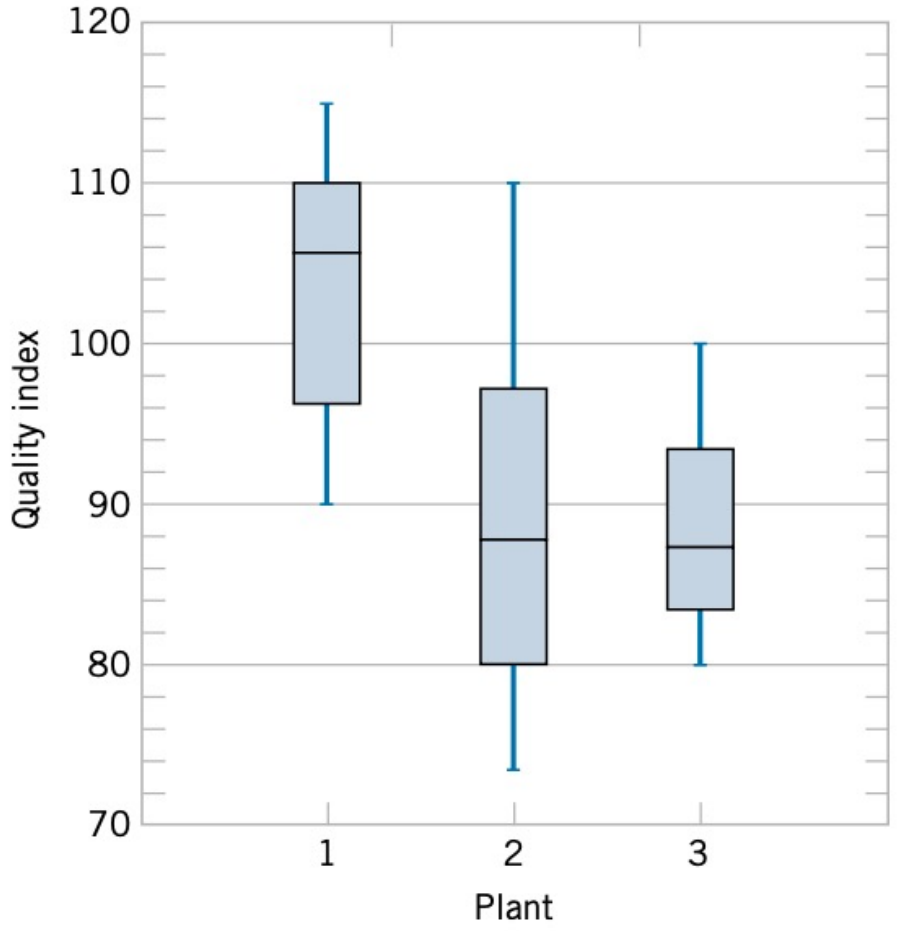
Box Plots


Whisker extends to smallest data point within 1.5 interquartile ranges from first quartile
Whisker extends to largest data point within 1.5 interquartile ranges from third quartile
First quartile
Second quartile
Third quartile
Outliers
Outliers
Extreme outlier
1.5 IQR
1.5 IQR
IQR
1.5 IQR
1.5 IQR

Comparison using box plot

Box plots are useful in graphical comparison of datasets

Comparative box plots of a quality index at three plants.



Time Series Plot

- A time series or time sequence is a data set in which the observations are recorded in the **order** in which they occur.
- A time series plot is a graph in which the vertical axis denotes the observed value of the variable (say x) and the horizontal axis denotes the time (which could be minutes, days, years, etc.).
- When measurements are plotted as a time series, we often see patterns like trends, cycles, or other broad features of the data

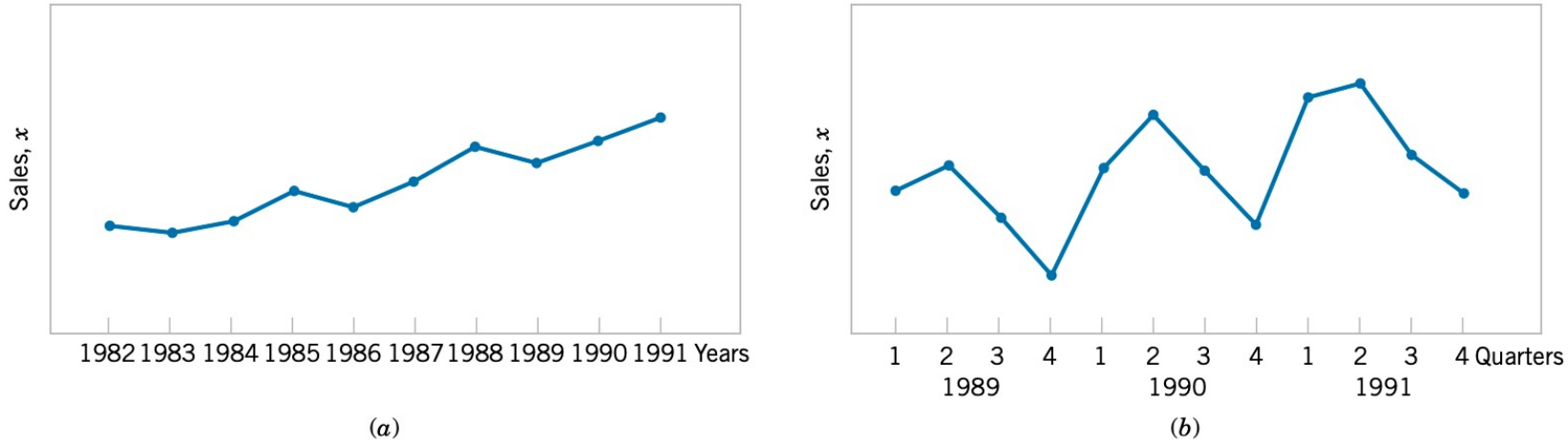


Figure: Company sales by year (a) and by quarter (b).

Normal probability plot

- A probability plot is a graphical method for determining whether sample data conform to a hypothesized distribution based on a subjective visual examination of the data.
- More reliable than the histogram for small- to moderate-size samples.
- Steps
  - Order the samples $x_{(1)}, x_{(2)}, \ldots, x_{(n)}$,
  - Ordered observations $x_{(j)}$ are plotted against the $(j - 0.5)/n$ quantile of the hypothesized distribution:

    $F(z_j) = (j - 0.5)/n$; F: CDF of hypothesized distribution
  - If hypothesized distribution (e.g., normal) is true, the plots will fall approximately along a straightline

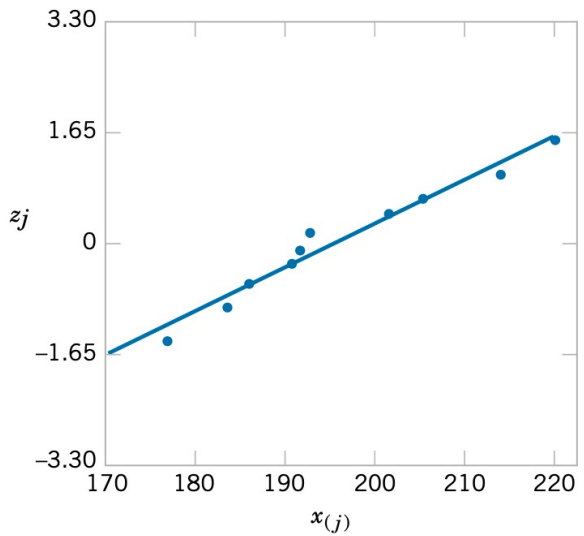
Example

A **normal probability plot** can also be constructed on ordinary graph paper by plotting the standardized normal scores z_j against $x_{(j)}$, where the standardized normal scores satisfy

$$\frac{j - 0.5}{n} = P(Z \leq z_j) = \Phi(z_j)$$

> Ten observations on the effective service life in minutes of batteries used in a portable personal computer are as follows: 176, 191, 214, 220, 205, 192, 201, 190, 183, 185. We hypothesize that battery life is adequately modeled by a normal distribution. To use probability plotting to investigate this hypothesis, first arrange the observations in ascending order and calculate their cumulative frequencies $(j - 0.5)/10$ as shown in Table 6-6.

j	$x_{(j)}$	$(j - 0.5)/10$	z_j
1	176	0.05	−1.64
2	183	0.15	−1.04
3	185	0.25	−0.67
4	190	0.35	−0.39
5	191	0.45	−0.13
6	192	0.55	0.13
7	201	0.65	0.39
8	205	0.75	0.67
9	214	0.85	1.04
10	220	0.95	1.64



Non-normal data

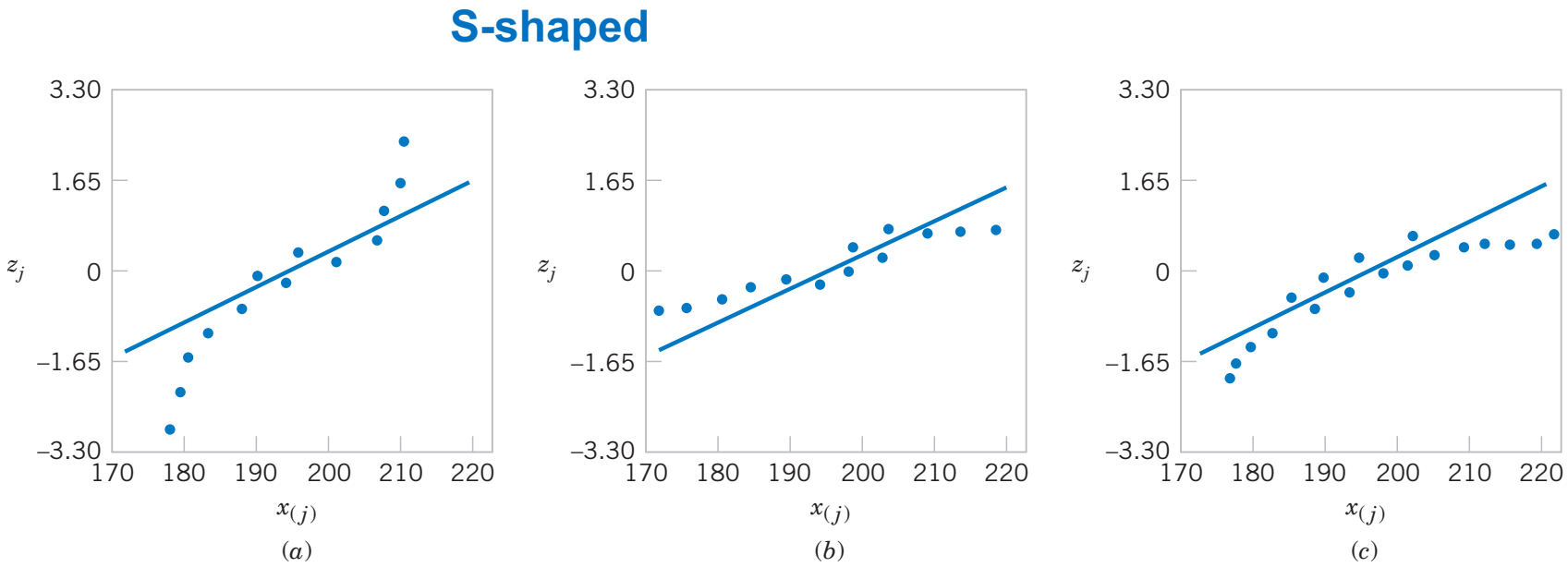
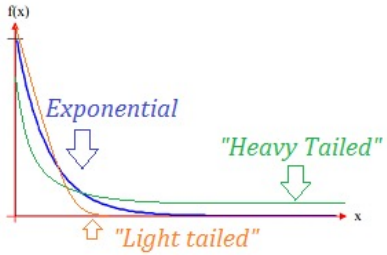
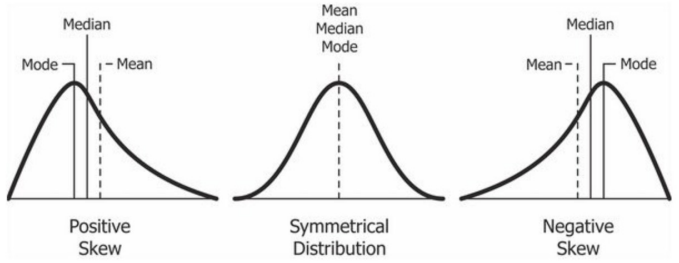


Figure 6-21 Normal probability plots indicating a nonnormal distribution. (a) Light-tailed distribution. (b) Heavy-tailed distribution. (c) A distribution with positive (or right) skew.

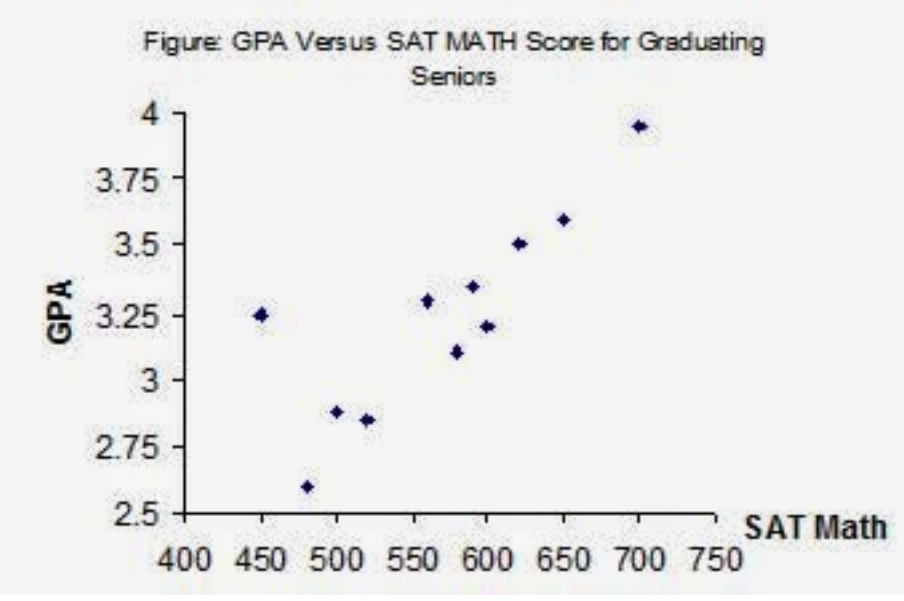
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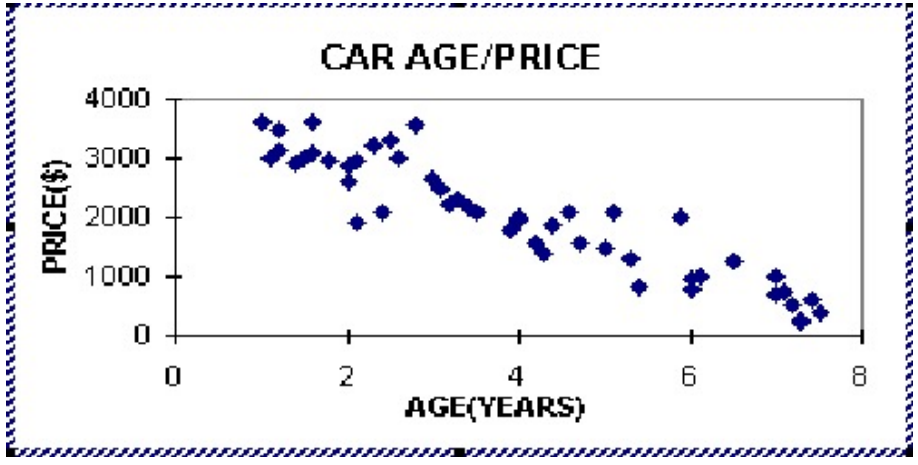




Scatter Diagrams

- In many problems, data scientists and statisticians work with **multivariate** data
- Scatter plot can graphically display the potential relationship between two variables
- When the two variables are linearly correlated, they should follow along a straight line



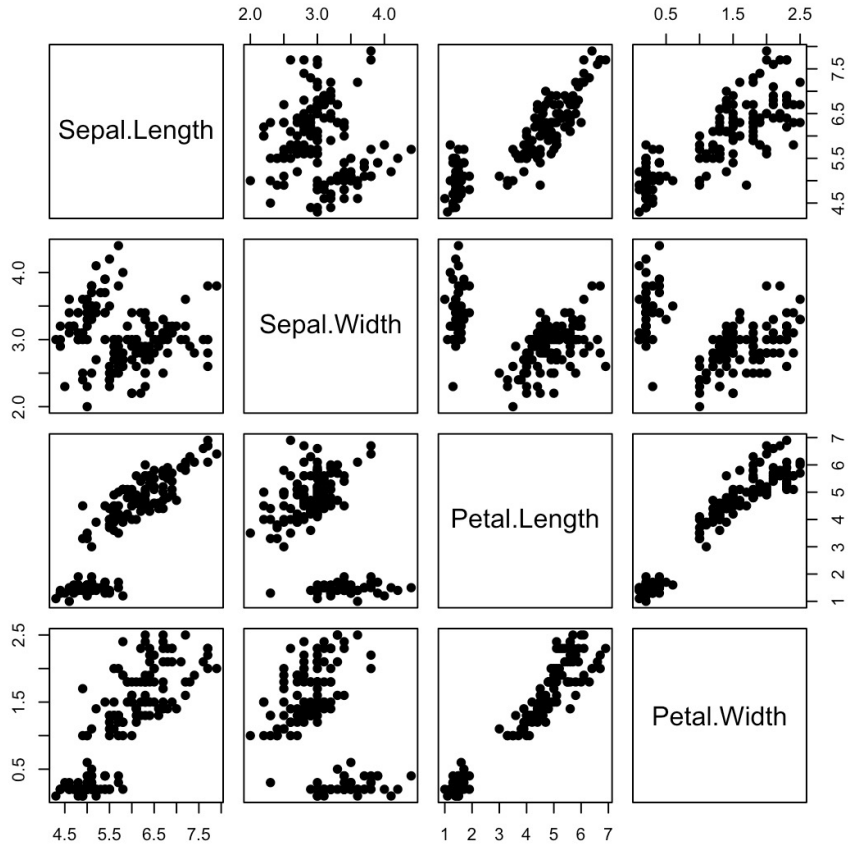


More than two variables

- When more than two variables are involved, can have a matrix of scattered diagrams







Example

Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

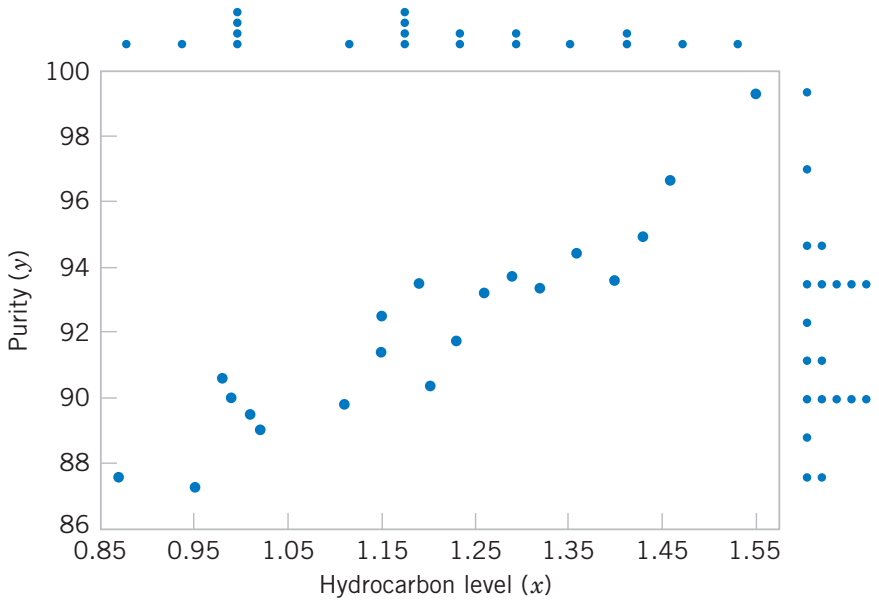
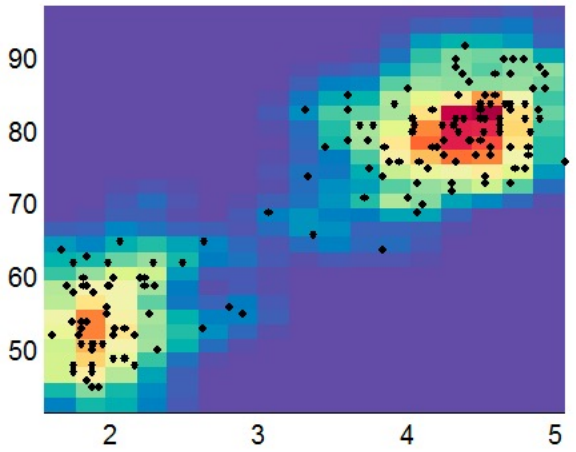
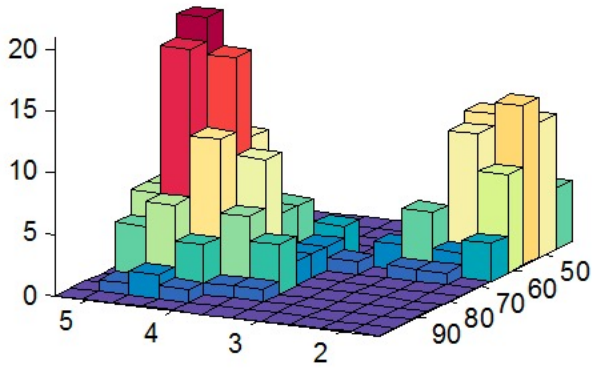
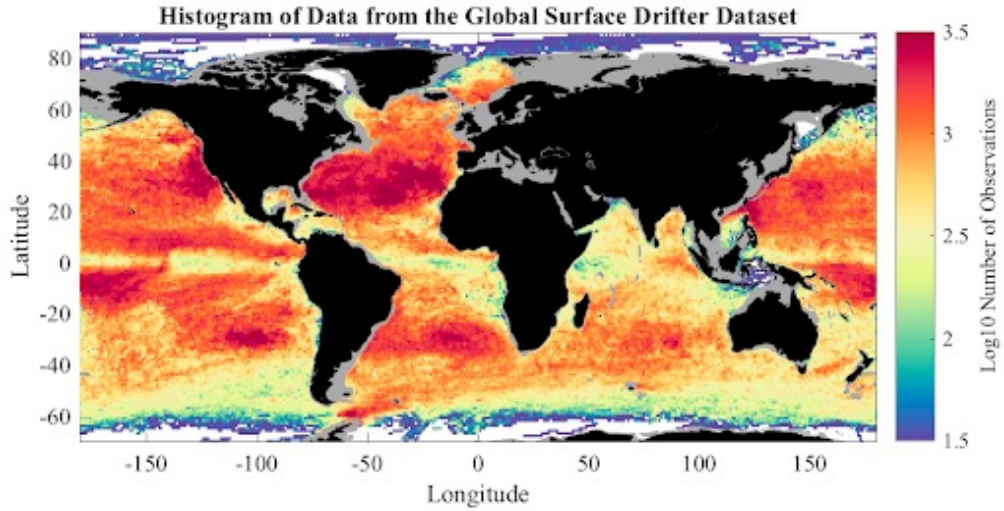


Figure: Scatter diagram of oxygen purity versus hydrocarbon level from Table 11-1.

Two Dimensional Histogram

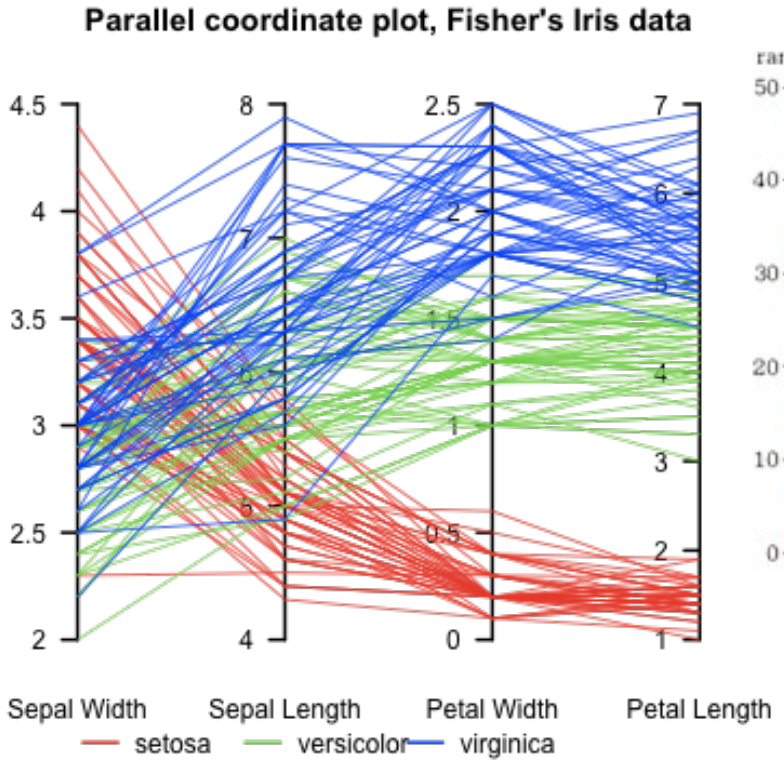


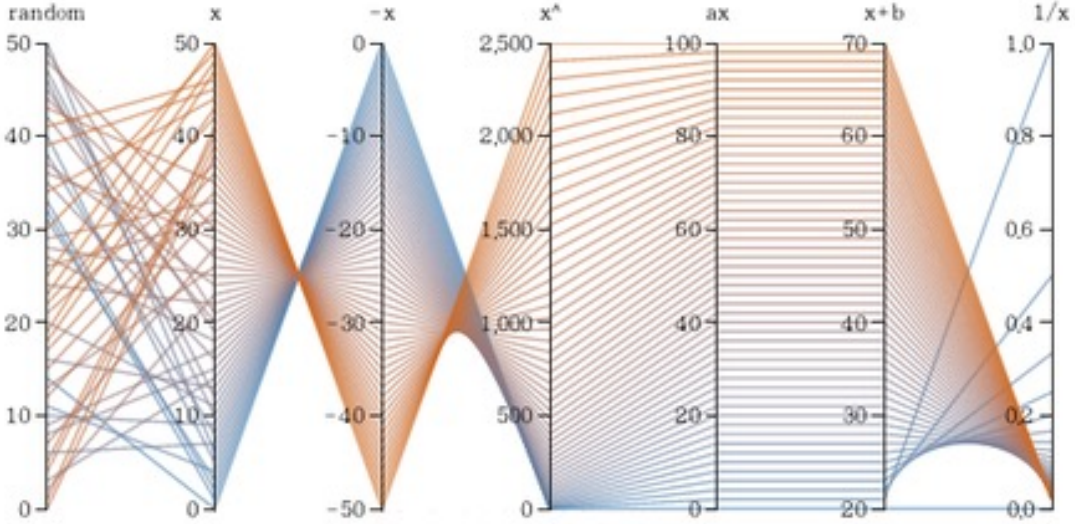




Parallel coordinates

- **Common way to visualize high-dimensional geometry and analyze multivariate data**







Iris Versicolor | Iris Setosa | Iris Virginica

Data visualization

- **https://ai.googleblog.com/2016/12/open-sourcing-embedding-projector-tool.html**

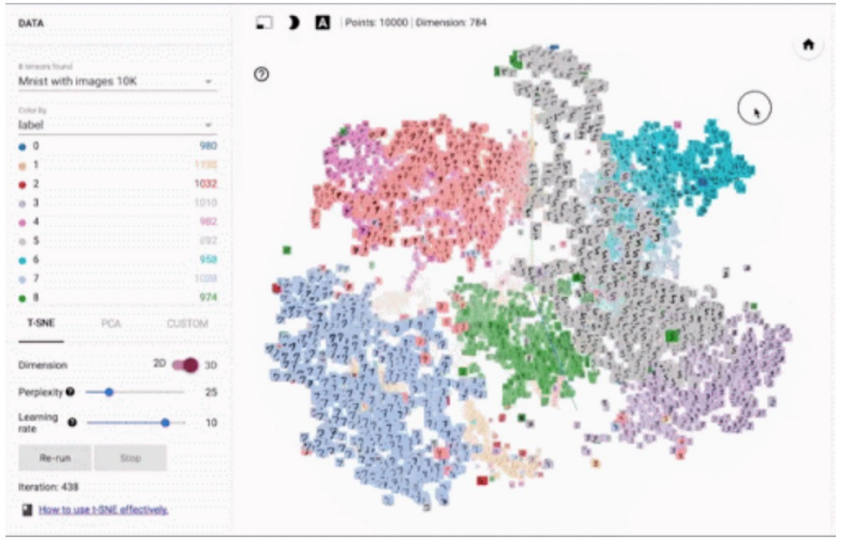
Google AI Blog

The latest news from Google AI

Open sourcing the Embedding Projector: a tool for visualizing high dimensional data

Wednesday, December 7, 2016

Posted by Daniel Smilkov and the Big Picture group



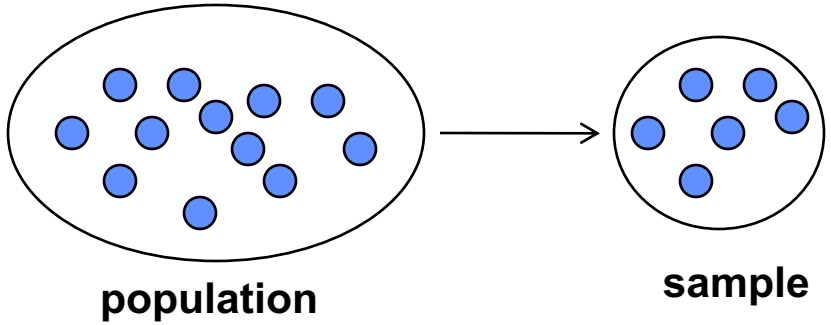
Descriptive Vs. Inferential Statistics

- Descriptive Statistics:

  A set of statistical techniques used to *organize*, *summarize*, *display*, and *describe* important features of data

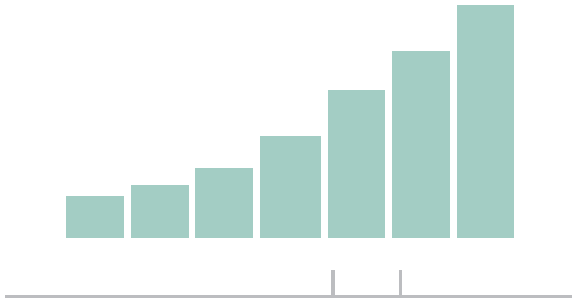
- Inferential (a.k.a. inductive) Statistics:

  A set of statistical methods that uses *sample* information to draw conclusion about the *population*



Exercise

1. Calculate the mean, median, mode, variance, IQR for the following data set {11, 2, 3, -1, 0, 2}. You can use either formula to calculate the variance.
2. Draw a box-plot of the above dataset.
3. Answer the following question by filling out the blank



The distribution is ____ skewed

Median ____ Mean